

$$\gamma_k(x_i) = \frac{\pi_k p(x_i | k)}{\sum_{j=1}^K \pi_j p(x_i | j)}$$

for each point x_i and each component k

shrink to 2 cases:
points x_1, x_2
components $k = 1, 2$, so $K = 2$

$$\gamma_1(x_1) = \frac{\pi_1 p(x_1 | 1)}{\pi_1 p(x_1 | 1) + \pi_2 p(x_1 | 2)}$$

$$\gamma_2(x_1) = \frac{\pi_2 p(x_1 | 2)}{\pi_1 p(x_1 | 1) + \pi_2 p(x_1 | 2)}$$

$$\gamma_1(x_2) = \frac{\pi_1 p(x_2 | 1)}{\pi_1 p(x_2 | 1) + \pi_2 p(x_2 | 2)}$$

$$\gamma_2(x_2) = \frac{\pi_2 p(x_2 | 2)}{\pi_1 p(x_2 | 1) + \pi_2 p(x_2 | 2)}$$

same denominator in each row: the two values of a row add up to 1